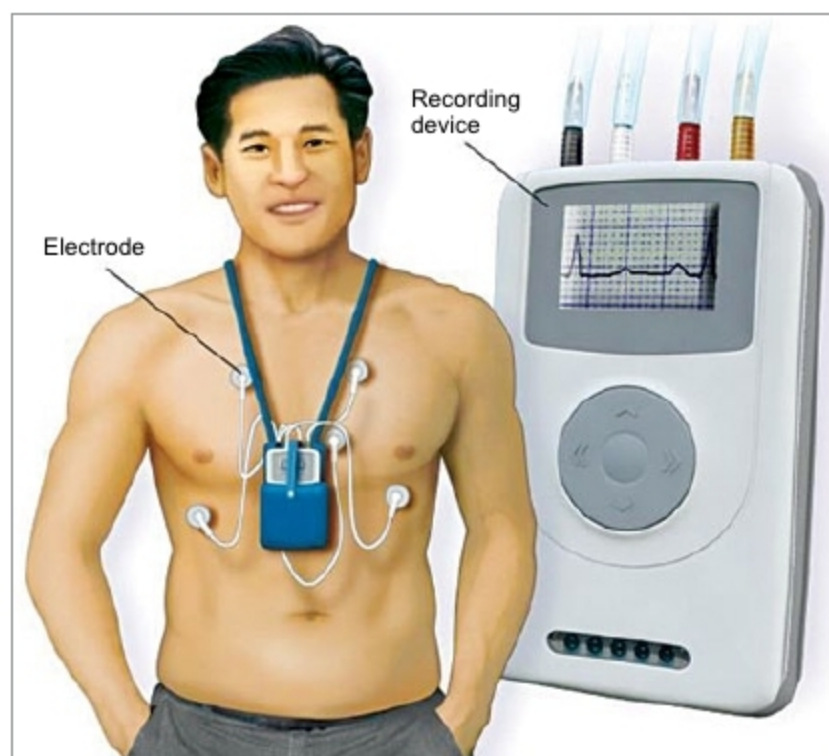
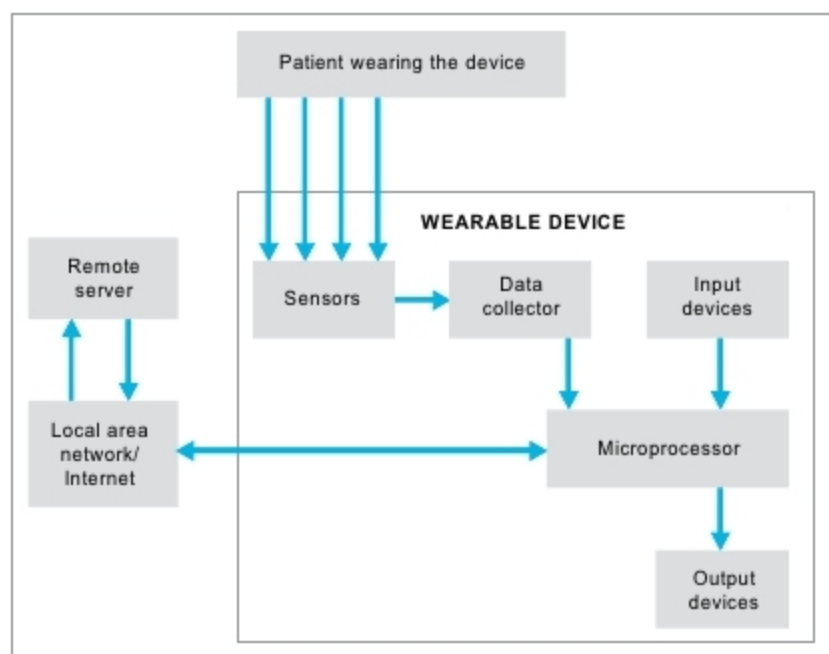




Heart holter, a wearable device that keeps track of the heart rhythm
(Credit: www.mayoclinic.org)



Layout of a wearable medical device

in a diary. This data is later analysed on a computer to correlate irregular cardiac activities, if any.

The need for obtaining real-time information and alerts led to the development of the next generation of wearable medical devices. These devices provide real-time wireless telemetry, feedback, medical alerts and clinical decisions by collecting, storing, processing and analysing clinical data. These have the ability to display and transmit information relating to health condition using non-invasive electronic, mechanical or biochemical sensors for monitoring physiological and kinesiological parameters. Advancements in micro sensor technology, computing technology, wireless communication capabilities and miniaturisation are assisting in the innovation

of advanced wearable medical devices.

Development of modern-day wearable devices was initially taken up for the US space programme when NASA needed to continuously monitor the essential medical parameters and physiological responses of astronauts in space. Further development was made by the US Army for developing wearable devices for use in battlefield with operational capabilities such as monitoring the soldier's health condition and his/her ability to fight.

A medical status monitor was used to read blood pressure, pulse, respiration and blood oxygen level, and transmit alarms along with the soldier's GPS location and severity of the injury. Later, improvements resulted in decision-making capability along with monitoring support.

Developments and improvements in the field are constantly

taking place to provide greater sophistication and facilities by using emerging technologies such as AI and nano-technology. Wearable medical devices are now freely available commercially for high-risk patients, athletes and for research purposes.

Main constituents of wearable medical devices

A wearable medical device receives inputs from sensors and the data collector. The microprocessor, which is the heart of the wearable medical device, provides data processing, data storage and wireless communication with the remote server. Sensors positioned on the body of the patient monitor him/her and the surrounding environment parameters, and feed this data to the data collector.

MULTI RANGE PROCESS INDICATOR

PI-48

Features

- 4 - Digit , 7 Segment LED Display
- DC Input 0 - 10V, 0 - 20mA & 4 - 20mA DC (User Selectable, All 3 Inputs in One Meter)
- Programmable Display -999 to 9999
- User Selectable Decimal Point
- High Accuracy : $\pm 0.5\%$ of FSD
- Auxiliary Supply : 85 - 265V AC / DC

Applications

- Process Industry
- Automation Industry
- Refinery Plant
- Variable Drives
- Process Equipments etc.

Measurement Range (DC)	Input Signal	Display
4-20mA	0 ~ < SLL*	- UL -
	SLL ~ 24mA	Measured Value
	> 24mA	- OL -
0-20mA	< 0mA	0000
	0 ~ 24mA	Measured Value
	> 24mA	- OL -
0-10V	< 0V	0000
	0 ~ 12V	Measured Value
	> 12V	- OL -

* Default SLL is 3.9mA (SLL : Set Low Limit)

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